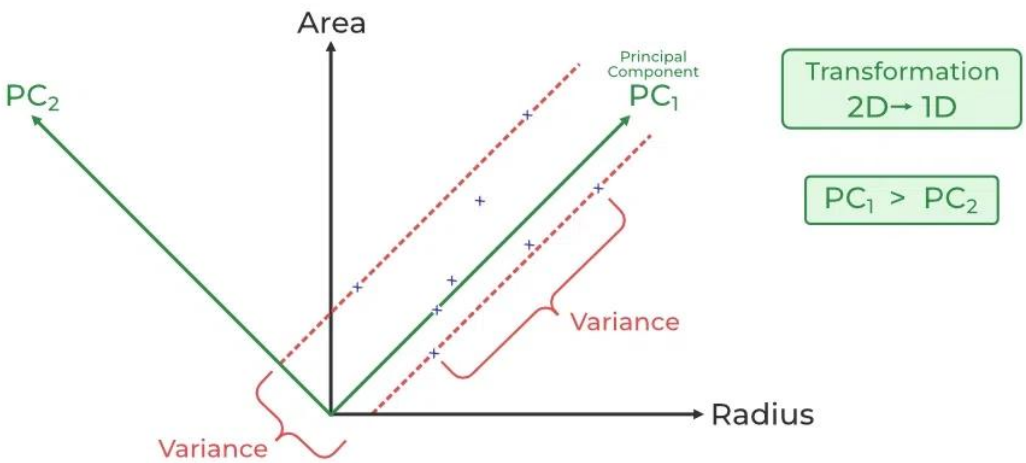
DAY 46

Principal Component Analysis (PCA)

1. What is PCA? (The Definition)

PCA is an unsupervised machine learning algorithm used for **dimensionality reduction**. It transforms a large set of variables into a smaller one that still contains most of the information (variance) from the original set.

Think of it like taking a 3D statue and finding the perfect angle to snap a 2D photo that captures every important detail.



2. Key Features of PCA

- **Dimensionality Reduction:** It reduces the number of variables in a dataset.
- **Feature Extraction:** It creates new variables (Principal Components) that are combinations of the old ones.
- **Orthogonality:** Each Principal Component (PC) is perpendicular to the others, meaning they are uncorrelated.
- **Variance Maximization:** The first PC captures the maximum possible variance in the data.

3. Why Do We Use It?

- **The Curse of Dimensionality:** Models struggle when there are too many features relative to the number of observations.
- **Visualization:** It's impossible to visualize 10D data. PCA lets us squash it into 2D or 3D.
- **Noise Reduction:** By keeping only the top components, we often filter out random "noise."
- **Efficiency:** Smaller datasets mean faster training times for your models.

4. How Do We Do It?

Standardization: Scale your data so every feature has a mean of 0 and a variance of 1. PCA is sensitive to variances; if one feature is measured in "kilometers" and another in "centimeters," the scale will bias the results.

1. **Covariance Matrix Computation:** Identify how variables vary from the mean with respect to each other (finding correlations).
2. **Eigen-Calculations:** Calculate the **Eigenvectors** and **Eigenvalues** of the covariance matrix.
    - **Eigenvectors:** Determine the *direction* of the new axes (the Principal Components).
    - **Eigenvalues:** Determine the *magnitude* or amount of variance captured by each direction.
3. **Feature Vector:** Choose whether to keep all components or discard the ones with low eigenvalues (low information).
4. **Recast the Data:** Project the original data onto the new principal component axes.